

A Hybrid Privacy-Preserving Framework for Federated Healthcare Analytics: Integrating Blockchain and Homomorphic Encryption for Secure Multi-Institutional Collaboration

Problem formulation:

Consider N number of healthcare institutions, where each institution i holds a private patient dataset D_i . Each dataset consists of multiple parameters (columns) that represent various patient attributes.

$D = \{D_1, D_2, \dots, D_n\}$ where D contains datasets of N healthcare institutions.

We formalize each dataset as:

$$D_i = \{D_i^{in}, D_i^{out}\}$$

where:

- Input factors: $D_i^{in} = \{(x_{i1}, x_{i2}, \dots, x_{im})\}$ represents n_i patient records with m features each (like age, blood pressure, glucose level, medical history)
- Output factors: $D_i^{out} = \{(y_{i1}, y_{i2}, \dots, y_{ik})\}$ represents the corresponding target labels or clinical outcomes (such as disease diagnosis, risk scores)

Each patient record can be denoted as (x_{ij}, y_{ij}) where $j \in \{1, \dots, n_i\}$.

Stage 1: Data Preparation

Preprocessing for compatibility with homomorphic encryption: mapping to polynomial rings in **CKKS** or integer vectors in **BFV** schemes

- Dataset $D_i = (D_i^{in}, D_i^{out})$ held by institution i , with input features $x_{ij} \in \mathbb{R}^m$ $j=1 \dots n_i$
- Labels $y_{ij} \in \{0, 1\}$

- If x is a feature in D_i^{in} , then $\text{Preprocess}(x) \rightarrow x'$, where x' conforms to the encoding requirements of the chosen HE scheme.
- Preprocess: Normalize and encode features for HE $x_{ij}' = \text{Encode}(\text{Normalize}(x_{ij}))$ so that for $\forall_{ij}, x_{ij} \rightarrow x_{ij}'$

Stage 2: Encryption

- Apply homomorphic encryption: $D_i^{enc} = \text{HE}(D_i^{in})$
- Homomorphic Encryption schemes: **CKKS** (for approximate arithmetic), **BFV/BGV** (for integer arithmetic), **TFHE** (for binary logic).
- Key generation: $(pk, sk) = \text{KeyGen}(1^\lambda, N, q)$.
- Ciphertext packing (**SIMD**): Enables efficient parallel computations

$\forall x_{ij} \in D_i^{in}, \text{HE.Encrypt}(pk, x_{ij}') \rightarrow c_{ij}$

$D_i^{enc} = \{c_{i1}, \dots, c_{ini}\}$

For batch processing: $\vec{c}_i = \text{HE.Encrypt}(pk, [x_{i1}', \dots, x_{inb}'])$

If D_i^{in} is pre-processed, then $\forall_j, (x_{ij}' \in D_i^{in}) \rightarrow (c_{ij} \in D_i^{enc})$

Stage 3: Local Training

- Each institution trains a local model: $M_i = \text{Train}(D_i^{enc}, D_i^{out})$
- If D_{new}^{enc} and M_{global} (encrypted or plaintext) exist, then $y^\wedge$ can be computed without accessing raw feature values.
- Key computations (forward pass, loss, backpropagation) performed homomorphically via polynomial approximations of activation functions (sigmoid, ReLU):

For encrypted input $\text{Enc}(x)$, weights $\text{Enc}(w)$, approximate σ by polynomial P :

$$\text{Enc}(y^\wedge) = P(\text{Enc}(w) \cdot \text{Enc}(x))$$

Gradient computation on ciphertexts:

$$\text{Enc}(\nabla L_i) = \frac{1}{n_i} \sum_{j=1}^{n_i} (\text{Enc}(\hat{y}_{i,j}) - y_{i,j}) \cdot \text{Enc}(x_{i,j})$$

Model update: $\text{Enc}(\Delta M_i) = \text{Enc}(w_i^{(t)}) - \text{Enc}(w^{(t-1)})$

Stage 4: Blockchain-Based Aggregation

- Institutions send encrypted model updates to blockchain-edge servers.
- Blockchain records each update as a transaction, timestamped and signed, ensuring tamper-proof tracking and auditability.
- Smart contracts verify the validity (format, signature) and reputation of clients before consensus aggregation.
- Consensus (delegated **PBFT**) is used to accept and aggregate valid encrypted updates (federated averaging or summation on ciphertexts).
- Encrypted updates $\text{Enc}(\Delta M_i)$ sent to blockchain edge servers with metadata and digital signatures.
- Smart contracts verify:

$$\text{Verify}(\text{Enc}(\Delta M_i), \sigma_i) \wedge \text{Reputation}(i) \geq \gamma$$

- Valid updates aggregated homomorphically:

$$\text{Enc}(\Delta M_{global}^{(t)}) = \sum_{i=1}^N \frac{n_i}{\sum_k n_k} \text{Enc}(\Delta M_i)$$

Stage 5: Prediction

Global model produces desired outputs: $\hat{y} = M_{global}(D_{new}^{enc})$

New encrypted input:

$$D_{new}^{enc} = \text{HE.Encrypt}(pk, D_{new}^{in})$$

Homomorphic inference using a global model:

$$\hat{y} = \text{HE.Eval}(M_{global}^{(t)}, D_{new}^{enc})$$

Optional trusted parties decrypt prediction:

$$y = \text{HE.Decrypt}(sk, \hat{y})$$

Integration Summary (Propositional Logic)

- For each institution i :



CKKS - Cheon-Kim-Kim-Song

PBFT - Practical Byzantine Fault Tolerance

BFV - Brakerski-Fan-Vercauteren

BGV - Brakerski-Gentry-Vaikuntanathan

SIMD - Single Instruction Multiple Data